An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 10: Fraud, Risk, and Trust & Safety

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Session plan (90 minutes)

1. Why fraud and trust & safety matter
2. Fraud and abuse categories
3. Fraud as a machine learning problem
4. Signals and features for fraud detection
5. Rule-based systems
6. Hands-on lab: risk scoring and thresholding

Learning objectives

By the end of this lecture, you should be able to:

- Identify major categories of fraud and abuse in e-commerce, including payment fraud, account takeover, promotion abuse, and refund abuse.
- Explain why fraud detection is a cost-sensitive, adversarial, and highly imbalanced decision problem.
- Compare supervised learning, anomaly detection, graph-based methods, and rule+ML hybrid approaches.
- Design a trust-and-safety workflow that combines scoring, thresholds, review operations, and business guardrails.

Why fraud and trust & safety matter

- payment costs,
- customer trust,
- seller trust in marketplaces,
- operational review burden,
- promotional efficiency,
- regulatory and compliance exposure.

Fraud and abuse categories

- **Payment fraud:** Payment fraud often involves stolen cards, synthetic identities, mule accounts, or coordinated abuse of payment flows.
- **Account takeover:** Account takeover occurs when an attacker gains control of an existing customer account.
- **Promotion and coupon abuse:** creating multiple accounts to redeem new-user offers,; coordinated coupon sharing,
- **Refund and return abuse:** Refund abuse includes false non-delivery claims, empty-box returns, repeated wardrobing behavior, and manipulation of return policies.
- **Marketplace and seller abuse:** counterfeit listings,; fake reviews,

Fraud as a machine learning problem

- **Class imbalance:** Fraudulent events are typically a small fraction of all transactions.
- **Adversarial behavior:** Once a platform changes a rule or scoring model, attackers may alter devices, payment instruments, shipping behavior, or timing patterns.
- **Label delay and label noise:** Ground truth often arrives late.
- **Asymmetric costs:** False negatives lose money and trust.

Signals and features for fraud detection

- **Transaction signals:** order value,; payment method,
- **Account signals:** account age,; prior successful orders,
- **Device and network signals:** device fingerprint,; IP reputation,
- **Behavioral and sequence signals:** login and checkout timing,; session depth,

Rule-based systems

- **Examples of useful rules:** block transactions from known bad cards or IPs,; trigger review if order value exceeds a threshold and account age is very low,
- **Strengths and weaknesses:** deterministic policy enforcement,; emergency response,

Supervised learning

- **Typical model families:** logistic regression,; tree-based ensembles,
- **Why supervised learning helps:** aggregating many weak predictors,; ranking risk more smoothly than rigid rules,
- **Operational caution:** the quality of labels,; the recency of training data,

Anomaly detection

- **What anomaly methods look for:** unusual transaction timing,; atypical device usage,
- **Strengths and weaknesses:** Anomaly methods are useful for surfacing suspicious patterns and complementing supervised systems.

Graph-based fraud analysis

- **Why graph structure matters:** many accounts linked to one device,; many cards shipping to related addresses,
- **Graph features and graph ML:** connected-component and degree features,; shared-entity counts,

Hybrid rules + ML systems

- **Why hybrids work:** enforce mandatory blocks,; assign risk scores,
- **Example decision flow:** apply deterministic policy rules,; compute risk score from supervised and anomaly signals,

Thresholds and decision economics

- **Approve, review, reject: Approve:** low-risk transactions proceed automatically.; **Review:** medium-risk cases go to human review or secondary checks.
- **Cost-sensitive tuning:** fraud loss,; customer lifetime value,

Human-in-the-loop review

- **What reviewers need:** transaction history,; device and network context,
- **Feedback loop:** rule refinement,; model retraining,

Fraud evaluation metrics

- **Useful metrics:** precision,; recall,
- **Business interpretation:** A model with strong recall but excessive false positives may block good customers.

Drift, adaptation, and monitoring

- **What to monitor:** score distribution shifts,; feature drift,
- **Why monitoring is essential:** fraud patterns shift,; product flows change,

Trust & safety beyond payment fraud

- **Examples beyond transactions:** review authenticity,; marketplace seller compliance,
- **Why this matters architecturally:** commerce events,; ERP refund records,

Hands-on lab: risk scoring and thresholding

- **Lab scenario:** Students are given a synthetic transaction dataset containing payment, account, device, and label information, plus estimated fraud-loss and false-positive costs.
- **Lab tasks:** Build a baseline risk model using structured transaction and account features.; Compute precision, recall, and a cost-sensitive evaluation summary.
- **Expected learning outcome:** By the end of the lab, students should be able to explain fraud detection as a staged decision system rather than a single prediction score.

Managerial implications

- clear policy,
- strong data quality,
- hybrid detection methods,
- review operations,
- and continuous monitoring.